

YANG-TIAN SUN

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🎓 EDUCATION

The University of Hong Kong

Oct 2023 – Present, Hong Kong

Ph.D. in Electrical and Electronic Engineering

Research interests: Video Generation; Generative Reconstruction; World (Action) Model

Supervisor: Prof. Xiaojuan Qi

University of Chinese Academy of Sciences

Apr 2019 – Jun 2022, Beijing, China

M.S. in Computer Science GPA 3.81/4.0

Research interests: Computer graphics, 3D computer vision, deep learning

Supervisor: Prof. Lin Gao

University of Science and Technology Beijing

Sep 2015 – Jun 2019, Beijing, China

B.S. in Applied Physics GPA 3.82/4.0 Ranking 1/86

📄 PUBLICATIONS

Publications (First Author / Co-First Author / Project Lead)

1. Mira-Scene: Pixel-Aligned Layouts for Generative 3D Scene

— “Aligns generated 3D scene layouts with image pixels for spatially coherent scene generation.”

Yang-Tian Sun*, Tianjia Liu*, Zehuan Huang, Yi-Hua Huang, Xiaoyang Lyu, Ziyi Yang, Zi-Xin Zou, Yuan-Chen Guo, Yan-Pei Cao[#], Xiaojuan Qi[#].

arXiv:2609.23796, 2026.

2. Stereo World Model: Camera-Guided Stereo Video Generation

— “Uses camera-guided stereo video generation to build geometrically consistent 3D-aware world models.”

Yang-Tian Sun, Zehuan Huang, Yiming Niu, Lin Ma, Yan-Pei Cao, Yuewen Ma, Xiaojuan Qi.

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

3. UniGeo: Taming Video Diffusion for Unified Consistent Geometry Estimation

— “Adapts video diffusion to estimate unified, temporally consistent scene geometry.”

Yang-Tian Sun, Xin Yu, Zehuan Huang, Yihua Huang, Yuan-Chen Guo, Ziyi Yang, Yan-Pei Cao, Xiaojuan Qi.

arXiv:2505.24521, 2025.

4. Splatter a Video: Video Gaussian Representation for Versatile Processing

— “Introduces a video Gaussian representation for versatile tracking, depth estimation, segmentation, and editing.”

Yang-Tian Sun*, Yi-Hua Huang*, Lin Ma, Xiaoyang Lyu, Yan-Pei Cao, Xiaojuan Qi.

The Conference and Workshop on Neural Information Processing Systems (NeurIPS), 2024, pp. 50401–50425.

5. SC-GS: Sparse-Controlled Gaussian Splatting for Editable Dynamic Scenes

— “Discovers sparse object controls for editable dynamic-scene reconstruction and manipulation.”

Yi-Hua Huang*, **Yang-Tian Sun***, Ziyi Yang*, Xiaoyang Lyu, Yan-Pei Cao, Xiaojuan Qi.

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

6. Robust Pose Transfer with Dynamic Details Using Neural Video Rendering

— “Separates dynamic appearance details from geometry to improve robust human pose transfer.”

Yang-Tian Sun, Hao-Zhi Huang, Xuan Wang, Yu-Kun Lai, Wei Liu, Lin Gao.

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 45(2), pp. 2660–2666, 2022.

7. Human Motion Transfer with 3D Constraints and Detail Enhancement

— “Combines explicit 3D constraints and detail enhancement for high-fidelity human motion transfer.”

Yang-Tian Sun, Qian-Cheng Fu, Yue-Ren Jiang, Zitao Liu, Yu-Kun Lai, Hongbo Fu, Lin Gao.

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 45(4), pp. 4682–4693, 2022.

8. **NeRF-Editing: Geometry Editing of Neural Radiance Fields**
— “Uses shape priors to enable interactive and semantically meaningful geometry editing of neural radiance fields.”
Yu-Jie Yuan*, **Yang-Tian Sun***, Yu-Kun Lai, Yuewen Ma, Rongfei Jia, Lin Gao.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

Publications (Co-Authored)

9. **Monocular Depth Estimation from a Single Image: Progress and Opportunities.**
Muxin Liu, Xiaoyang Lyu, **Yang-Tian Sun**, Yihua Huang, Ziyi Yang, Peng Dai, Xiaojuan Qi.
arXiv:2609.01172, 2026.
10. **Stabilizing Streaming Video Geometry via Dynamic Feature Normalization.**
Xiaoyang Lyu, Mingjie Liu, Xiuzhe Wu, Rui Wang, Yihua Huang, **Yang-Tian Sun**, Shiyi Shi, Xiaojuan Qi.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
11. **Repurposing 3D Generative Model for Autoregressive Layout Generation.**
Haoran Feng, Yiming Niu, Zehuan Huang, **Yang-Tian Sun**, Chenyang Guo, Yifan Peng, Lu Sheng.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
12. **VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory.**
Yifei Yu, Xiaoshan Wu, Xinting Hu, Tao Hu, Yangtian Sun, Xiaoyang Lyu, Bo Wang, Lin Ma, Yuewen Ma, Zhongrui Wang, Xiaojuan Qi
arXiv:2512.04519, 2025.
13. **SeqTex: Generate Mesh Textures in Video Sequence.**
Ze Yuan, Xin Yu, **Yang-Tian Sun**, Yuan-Chen Guo, Yan-Pei Cao, Ding Liang, Xiaojuan Qi.
Proceedings of the SIGGRAPH Asia 2025 Conference
14. **AnimaX: Animating the Inanimate in 3D with Joint Video-Pose Diffusion Models.**
Zehuan Huang, Haoran Feng, **Yang-Tian Sun**, Yuanchen Guo, Yanpei Cao, Lu Sheng.
Proceedings of the SIGGRAPH Asia 2025 Conference
15. **Deformable Radial Kernel Splatting.**
Yi-Hua Huang, Ming-Xian Lin, **Yang-Tian Sun**, Ziyi Yang, Xiaoyang Lyu, Yan-Pei Cao, Xiaojuan Qi.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
16. **Spec-Gaussian: Anisotropic View-Dependent Appearance for 3D Gaussian Splatting.**
Ziyi Yang, Xiaoyang Gao, **Yang-Tian Sun**, Yi-Hua Huang, Xiaoyang Lyu, Wen Zhou, Shibo Jiao, Xiaojuan Qi, Xiaogang Jin.
Neural Information Processing Systems (NeurIPS), 2024, pp. 61192–61216.
17. **3DGSr: Implicit Surface Reconstruction with 3D Gaussian Splatting.**
Xiaoyang Lyu, **Yang-Tian Sun**, Yi-Hua Huang, Xiuzhe Wu, Ziyi Yang, Yilun Chen, Jiangmiao Pang, Xiaojuan Qi.
ACM Transactions on Graphics (TOG), 43(6), 2024.
18. **3D-Aware Text-Driven Talking Avatar Generation.**
Xiuzhe Wu, **Yang-Tian Sun**, Handi Chen, Hang Zhou, Jingdong Wang, Zhengzhe Liu, Xiaojuan Qi.
European Conference on Computer Vision (ECCV), 2024, pp. 416–433.
19. **Total-Decom: Decomposed 3D Scene Reconstruction with Minimal Interaction.**
Xiaoyang Lyu*, Chirui Chang*, Peng Dai, **Yang-Tian Sun**, Xiaojuan Qi.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
20. **Interactive NeRF Geometry Editing with Shape Priors.**
Yu-Jie Yuan, **Yang-Tian Sun**, Yu-Kun Lai, Yuewen Ma, Rongfei Jia, Leif Kobbelt, Lin Gao.
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 45(12), 2023.

 HONOURS AND AWARDS

- **National Scholarship**, 2015/2016 academic year
- **National Scholarship**, 2016/2017 academic year

⚙️ WORKING EXPERIENCE

VAST

Jul 2023 – Now

Research Intern

- 3D/4D (Generative) Reconstruction, World Model, Real2Sim

NetEase Game AI Lab

Jul 2022 – Sep 2023

AI Engineer

- Applied differentiable rendering to game asset production.

Alibaba Taobao

Jul 2021 – Oct 2021

Research Intern

- Investigated deformation algorithms for neural radiance fields.

Tencent AI Lab

Jun 2020 – Mar 2021

Research Intern

- Investigated dynamic-detail generation for neural human rendering.

♥️ PROFESSIONAL SERVICES

Reviewer of SIGGRAPH, SIGGRAPH Asia, T-PAMI, CVPR, ICCV, ECCV, ICLR, NeurIPS, TVCG